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2 authors:

Tarun Gupta
Indian Institute of Technology Kanpur

Anand Kumar
University of British Columbia - Vancouver

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Development and Field Evaluation of a Multiple Slit Nozzle-Based High Volume $PM_{2.5}$ Inertial Impactor Assembly (HVIA)

Anand Kumar[a,b] and Tarun Gupta[a,b]

[a]*Environmental Engineering and Management Program,* [b]*APTL CESE*

Indian Institute of Technology Kanpur, Kanpur - 208016 (India)

HIGHLIGHTS

- Development of a multiple slit nozzle-based high volume $PM_{2.5}$ inertial impactor assembly.
- Validation of the impactor assembly with an established low flow rate $PM_{2.5}$ impactor.
- Comparison of $PM_{2.5}$ concentration and chemical composition obtained from 2 impactors.

ABSTRACT

This study presents the development and lab performance of a high volume (Q = 950 LPM), multiple slit nozzle-based $PM_{2.5}$ (particle aerodynamic diameter < 2.5 µm) inertial impactor and the performance of the same was compared with an existing single stage low flow rate (Q = 15 LPM) $PM_{2.5}$ impactor. Lab experiments were performed on various slit-based nozzle impactors using polydisperse dolomite powder as test aerosol. After carrying out rigorous parametric evaluation, the optimum slit nozzle-based impactor configuration selected had cutoff size of 2.51 µm (aerodynamic diameter) at an operating flow rate of 215 LPM (medium flow) with a pressure drop of 0.35 kPa across the impactor stage. The length of the slit of the optimum medium flow impactor was extrapolated to a flow rate of 950 LPM to obtain the high volume multiple slit nozzle-based $PM_{2.5}$ inertial impactor assembly. This novel impactor assembly was fabricated from brass and chrome-plated and then retrofitted in a high volume dust sampler (Model AAS 217NL, Ecotech Instruments, India) downstream of the PM_{10} cyclone separator. High vacuum silicone grease was used as the impaction substrate. A field study was performed with co-located novel high volume impactor assembly (HVIA) and single stage low flow rate $PM_{2.5}$ impactor, to not only compare the $PM_{2.5}$ mass concentrations but elemental, anion and water soluble organic carbon (WSOC)/ water soluble inorganic carbon (WSIC) concentrations as well in order to validate the HVIA developed in the present study.

Keywords: Impactor, $PM_{2.5}$, Slit-Nozzle, Silicone Grease.

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[b]***Corresponding author, Tel: +91-512-2597128; E-mail address: tarun@iitk.ac.in***

INTRODUCTION

$PM_{2.5}$ (particles having aerodynamic diameter < 2.5 μm as defined in human health-related studies; while based on USEPA convention for size separator, these are particles which pass through a size-selective inlet with a 50% efficiency cut-off at 2.5 μm aerodynamic diameter) has received special consideration as various epidemiological and toxicological studies have shown substantial evidences of direct harmful health impacts of $PM_{2.5}$ on the human population. Studies have shown correlation between mass concentration of $PM_{2.5}$ and health effects in sensitive population (elderly and diseased) and definite functional changes in the human cardiovascular and respiratory system (Pope et al., 2011).

Exposure to $PM_{2.5}$ has been associated with changes in morbidity and mortality rates, with changes in respiratory function, and with cardiovascular dysfunctions in numerous epidemiological studies (Schwartz and Dockery, 1992; Korrick, 1998; Neas, 1999). For the same amount of particulate matter (PM) mass deposited in the lung, toxicity tends to increase as particle size decreases. This is attributed to the increased surface area per unit mass available or to the ability of finer particles to penetrate the lung tissues and subsequently enter into the blood stream (Schwartz et al., 1996; Harrison and Yin, 2000; Cohen et al., 2005; Kunzli and Tager, 2005; Sharma and Agrawal, 2005; Huang and Ghio, 2006). Hence $PM_{2.5}$ is considered as a better exposure indicator than PM_{10} for health impact assessments (WHO, 2006). Conducting studies to investigate the acute, chronic, and subchronic health effects of such particles indeed becomes imperative. As a result, it is necessary to collect relatively large amounts of particles in this size range which requires air samplers operating at substantially higher flow rates.

Inertial impactors are relatively simple devices, based on the concept of impaction where air laden with particle flows around an impaction substrate and is subjected to sharp change in air flow trajectory. Particles with sufficient inertia will slip across the sharply bending air streamlines and impact on the impaction surface while the finer particles will follow the bending air streamlines and move downstream of the impaction substrate (Hinds, 1999). The finer particles could be collected on a suitable filter paper placed downstream of the impactor and it could be analyzed gravimetrically and chemically later on. Sampling high volume of air with an inertial impactor is advantageous but it also projects a lot of problems in terms of flow leakage, impaction substrate overloading and particle bounce off. A number of round nozzle-based impactors are available for $PM_{2.5}$ speciation (Marple et al., 1991; Demokritou et al., 2001; Peters et al., 2001), but not many are there with rectangular slit nozzle-based. Slit-based nozzles provides lower jet Reynolds number and pressure drop compared to round nozzle impactors (Kumar and Gupta, 2014).

This study presents the development, lab performance and field evaluation of a user-friendly high volume (Q = 950 LPM), multiple slit nozzle-based $PM_{2.5}$ inertial impactor assembly that can effectively counteract the problems of substrate overloading and particle bounce off. Gravimetric-based field comparison of the developed high volume sampler with a previously established low flow rate single stage $PM_{2.5}$ sampler was also carried out to validate the accuracy of the novel sampler.

METHODOLOGY

Design Considerations for the Impactors

Our group has developed other low flow rate impactors as well in the past (Gupta et al., 2010; Gupta et al., 2011; Kumar and Gupta, 2014). Similar design considerations and methodology, as used in designing previous impactors, was adopted in developing this novel slit-based impactor. The cutoff size of an impaction stage can be calculated by using the Stokes equation, as follows (Hering, 1995; Baron and Willeke, 2001):

$$Stk = \frac{U\, C_c\, \rho_p\, d_p^2}{9\, \eta\, W} \qquad [1]$$

Where, ρ_p is the particle density (kg/m^3), η is the dynamic viscosity of air (Pa.s), d_p particle diameter (m), U jet velocity in the impactor nozzle (m/s), W width of slit nozzle (m), and C_c Cunningham slip correction factor. The theoretical value of $\sqrt{Stk_{50}}$ was taken as 0.70 for the design of slit nozzle (Marple and Willeke, 1976). The nozzle Reynolds number (Re) was calculated using the following equation:

$$Re = \frac{2\, U\, W \rho_{air}}{\eta} \qquad [2]$$

Where ρ_{air} is the air density (kg/m^3), U is the flow velocity (m/s) at standard conditions.

Description of the Experimental Setup

The experimental setup used for analyzing impactor characteristics is shown in the Fig. 1. The experimental setup consisted of three components: the aerosol generation system, the dilution chamber and the particle monitoring system. An improved dry aerosol generator (the available aerosol generator was modified by making design changes in the powder dispersion nozzle and it tapering angle to improve the performance of the same in terms of aerosol size and generation stability) was employed using dolomite powder to produce a stable flow of polydisperse aerosol. The stability of the aerosol generation was maintained using needle valves that directly controlled the flow of compressed air entering the aerosol chamber. The dolomite powder used for generating dry aerosol had particle density as 2840 kg/m^3 and shape factor as 1.10 (Deelman, 1999; Alkuwairan, 2012). The powder had been sieved through 45 µm mesh and the finer fraction was used for aerosol generation in order to avoid clogging of injector nozzle. The generated aerosol enters the mixing chamber where it was mixed and diluted with dried, HEPA (High Efficiency Particulate Filter) filtered ambient air. Then the diluted aerosol enters into a cylindrical duct of approximately 2 m length on the upstream side of impactor assembly to ensure complete mixing and uniform concentration of aerosol at the sampling point. A high flow rate vacuum pump was employed to generate the desired flow rate. This flow rate was monitored by a rotameter previously calibrated with a mass flowmeter (Model 4040, TSI Inc., USA).

A portable aerosol spectrometer (OPC 1.108, Grimm GmbH, Germany) was used to test the performance of the impactor. Since OPC gives optical diameter (d_0) of the particles, so Eq. (3) (Hinds, 1999) was used to convert it to aerodynamic diameter (d_a):

$$d_a = d_o \left[\frac{\rho_p}{\chi \rho_0}\right]^{1/2} \qquad [3]$$

Design, Fabrication and Lab Evaluation of a Multiple Slit Nozzle-based Medium Flow PM$_{2.5}$ Impactor

Various slit-based impactor nozzles, with width in the range of 1.5 mm and 6 mm, were designed for PM$_{2.5}$ based on the Eq. (1) – (2) keeping the flow rate in the range of 100 – 225 LPM. Impactor plates were fabricated keeping appropriate spacing between the nozzles to avoid jet-to-jet interactions (Fang et al., 1991). The impactor nozzles along with impaction substrate units were tested individually using the laboratory testing rig setup. High vacuum silicone grease (Dow

Corning, USA) was used as the impaction substrate in all cases. To minimize the bounce off and breakup losses of large particles, a 4 mm thick and smooth layer of grease was created by using a razor blade (Turner and Hering, 1987; Park et al., 1992; Hill et al., 1999). OPC 1.108 was used to alternatively sample between upstream and downstream of the impactor for a time interval of 4 min each. The difference in particle concentration was then calculated for the upstream and downstream flows to determine the collection efficiency of the impactor nozzle. Each experiment consisted of at least 4 sets of upstream and downstream samples. In addition, each configuration was repeatedly tested for at least 2-3 times. The optimized nozzle configuration had 2 mm wide slit-type accelerating nozzles (total length = 70 mm) with a cutoff size 2.51 μm at an operating flow rate of 215 LPM.

Design and Fabrication of a Multiple Slit Nozzle-based High Volume $PM_{2.5}$ Impactor Assembly (HVIA)

Finally, the optimum multiple slit-based nozzle configuration (parent assembly) obtained for $PM_{2.5}$ (Q = 215 LPM) in the previous section, was extrapolated to a flow of 950 LPM (final slit length of HVIA = (950/215)*70 = 309.30 mm). Essentially, all the impactor characteristics remained the same as that of the medium flow impactor except that the slit length was increased in proportion with the flow rate. Hence, no impactor characterization was needed to be done on HVIA, rather, length of parent assembly was increased to obtain the HVIA. The smoothness of inner walls of the nozzle is an important factor in reducing particle losses. Therefore, impactor nozzles were fabricated from brass (and chrome-plated after fabrication) as it was easier to machine according to the design specifications and provided better finishing. The impactor assembly was retrofitted in the commercially available high volume dust sampler (Model AAS 217NL, Ecotech Instruments, India) downstream of the PM_{10} cyclone separator. High vacuum silicone grease as used in the previous impactors was employed with this impactor as well to eliminate particle bounce-off.

The optimum medium flow parent slit impactor (Slit Length = 70 mm; Q = 215 LPM) was also tested at flow rates of 205 LPM (corresponding to 900 LPM) and 225 LPM (corresponding to 1000 LPM) to get an idea of the change in cutoff over operating flow range of the dust sampler. The impactor nozzle was attached to sampler body with the help of screws along the air duct on the downstream of the cyclone. Interlocking spacer provided with the nozzle plate ensures proper orientation of the substrate plate and maintains the desired S/W. Fig. 2(a) and 2(b) shows the schematic and various components of the novel slit assembly, respectively.

Sampling Site and Schedule

Field evaluation of the high volume dust sampler retrofitted with HVIA was carried out inside IIT Kanpur campus premises between 11th July 2014 and 31st August 2014 for 30 days. IIT Kanpur (26.43°N, 80.33°E) is an academic and research institute with residential facilities and no major industrial and/or commercial activities. Sampling was carried out (for 8 hours on each sampling day) with co-located dust sampler retrofitted with HVIA developed in the current study and a previously established single stage low flow rate $PM_{2.5}$ (single round nozzle type) impactor (Gupta et al., 2011) operating at 15 LPM. Both samplers were placed 4 m apart in a well ventilated corridor in front of Room 203, Centre for Environmental Science & Engineering (CESE), IIT Kanpur, at a height of 5 m from ground level. Teflon filters (PTFE, Pall Life Sciences, Teflon w/ring, PTFE membrane, porosity 2.0 µm) of 47 mm diameter were used with low flow rate impactor to collect fine particles while A4 size (8” X 10”) Glass Microfibre filters (EPM 2000, Whatman) were used with HVIA.

Sample Extraction and Chemical Analysis

All filters were pre-conditioned by equilibrating it for 24 hours at 21 ± 2°C (room temperature) and relative humidity (35 ± 3%) (Stone et al., 2011). The preconditioned 47 mm diameter filters were weighed using micro weighing balance (Mettler Toledo MT5, with a resolution of 1 μg) while A4 size filters were weighed using 5-digit lab grade balance (Mettler Toledo, Least Count = 0.1 mg). After sampling, the filters were subjected to post conditioning as well in order to eliminate errors in gravimetric weighing which can be introduced due to the presence of moisture. After post-conditioning, filters were weighed to obtain $PM_{2.5}$ mass concentration.

Out of the 30 sets of 47 mm diameter Teflon filters and A4 size filters, 15 sets were used to carry out the chemical analysis. Each 47 mm diameter filter was cut into 2 halves with the help of plastic scissors and forceps. One half of the filter paper was subjected to acid digestion (HNO_3 acid 65%, GR Merck Supra pure) and 100 ml filtered aliquots were prepared following previously established method (Chakraborty and Gupta, 2010) to analyze them for elements using ICP-OES (Inductively Coupled Plasma-Optical Emission Spectroscopy, ICAP 6300 Thermo Inc.). Other half of the filter paper was subjected to aqueous extraction in an ultrasonic bath for 30 minutes (Fast Clean Ultrasonic cleaner, 2k909008) and 100 ml filtered aliquots were prepared to analyze for 3 anions (Cl^-, NO_3^- and SO_4^{2-}) using Ion Chromatography (882 Compact IC plus, Metrohm Inc., Switzerland) as per the previously established method (Gupta and Mandaria, 2013). The filtered samples were also analyzed for water soluble organic carbon (WSOC) and inorganic carbon (WSIC) using TOC Analyzer (TOC-V CPN, Shimadzu Corporation). In case of A4 size filters, similar extraction procedure was adopted for element, anion and WSOC/WSIC analysis by using $1/8^{th}$ of the filter paper. Validation of the HVIA was done by comparing the $PM_{2.5}$ mass concentrations and the concentration of the chemical species obtained from the two samplers. Linear correlation coefficients and student t-Test were also conducted on the mass concentration and chemical analysis data to establish non-significantly different performance of the two samplers.

RESULTS AND DISCUSSION

Parametric Investigations and Impactor Characterization for Multiple Slit Nozzle-based Medium Flow $PM_{2.5}$ Impactor (Parent Assembly)

Various slit-based nozzles were designed and rigorously tested with the laboratory testing rig. Table 1 presents the details of the initial experiments performed on various slit-based impactor nozzles and summarizes the results from the parametric investigations carried out before arriving at an optimum configuration. The last column gives a comparative view of the sharpness of collection efficiency curves obtained from the investigation. In real impactors, the jet flow velocity is uniform in the central region of the slit nozzle but reduces to zero at the nozzle wall. Such varying velocity profile affects the sharpness of the efficiency curve. To analyze such effect, nozzles with smaller width (upto 1.5 mm) were also tested. The collection efficiency curves obtained for the tests performed on various impactor nozzles are shown in Fig. 3. After analyzing and comparing the relevant parameters (cutoff size, sharpness of collection efficiency curve, $\sqrt{Stk_{50}}$ and pressure drop across impactor) obtained through the rigorous experimentation, an optimum $PM_{2.5}$ configuration was selected which had 2 mm wide slit-type accelerating nozzles (total length = 70 mm) with an experimental cutoff of 2.51 µm at an operating flow rate of 215 LPM. The nozzle losses obtained for the optimized impactor nozzle is also shown in Fig. 3 over the secondary vertical axis. The losses were well below 10% for particle diameter less than 3 µm. The sharpness (geometric standard deviation σ_g which is the square root of the ratio d_{84}/d_{16}; d_{84}

and d_{16} are particle diameter corresponding to particle collection efficiencies of 84% and 16%, respectively) of the collection efficiency curve was 2.14, which is on the higher side. This can be attributed to the limited number of particle size bins available in the OPC. Farther spaced diameter data channels (0.3, 0.4, 0.5, 0.65, 0.8, 1, 1.6, 2, 3, 4, 5, 7.5, 10, 15, 20 μm) available from OPC does not allow the collection efficiency plot to be as steep as it could have been in case of closely-spaced diameter channels (e.g. in aerodynamic particle sizers). Pressure drop was 0.35 kPa (1.4 inches of H_2O) which was fairly low. Comparing these results with the study conducted by Kumar and Gupta (2014) on multiple round nozzle $PM_{2.5}$ impactor shows that lower Reynolds number and pressure drop can be obtained with slit nozzles at similar flow rates. Hence, slit-based nozzles provide an upper hand over round nozzles.

The degree of particle re-entrainment is a function of the type of particle (size, shape and density) and the nature of the impaction surface (Rao, 1975). The impactor characterization was carried out using dolomite powder as test aerosol with particle density of 2840 kg/m^3, which is comparable to the density of mineral dust aerosol. Fig. 4 shows average aerosol number concentration on the upstream and downstream of the impactor in one of the parameterization tests at 215 LPM which usually takes an hour for completion. Particles with size greater than 4 μm are very low in number on the downstream, which shows that particle bounce-off and re-entrainment were very low even with such test aerosol. In addition, thick and smooth silicone grease coating of 0.20 cm or greater can help avoid the problem of particle bounce and grease can withhold the collected particles on the surface more effectively (Hill et al., 1999). This kind of impaction substrate was found to be very effective in eliminating particle bounce-off as particles were rather deeply embedded into the grease and silicone oil wetted their surface soon after impaction onto the grease substrate (Turner and Hering, 1987; Demokritou et al., 2001).

Retrofit Dust Sampler with Multiple Slit Nozzle-based High Volume $PM_{2.5}$ Impactor Assembly (HVIA)

The parent assembly serves as a small-scale version of HVIA, and all parameterization tests were performed on the parent assembly only. The optimum parent assembly obtained earlier (as described in the previous section) was extrapolated to a flow rate of 950 LPM (in the operating flow range of the dust sampler Model AAS 217NL, Ecotech Instruments, India) to achieve a cutoff of 2.51 μm (final slit length = (950/215)*70 = 309.30 mm) with similar collection efficiency curve, $\sqrt{Stk_{50}}$ and pressure drop as of the parent medium flow rate slit impactor. Hence, no impactor parameterization was needed to be done in the laboratory on HVIA. The HVIA was installed on the downstream of the cyclone separator (for removing particles of size greater than 10 μm) present in the dust sampler. Fig. 5 shows the components of the dust sampler with the novel HVIA installed on it. Particle loading on the substrate is substantially reduced and its performance is enhanced since particles of size greater than 10 μm are removed beforehand (by the cyclone).

Considering the operating flow range of the dust sampler and the possible reduction in the working flow rate due to the pressure drop introduced because of the impactor assembly, the parent assembly (medium flow slit-based impactor parent assembly) was tested accordingly at flow rates of 205 LPM (corresponding to 900 LPM) and 225 LPM (corresponding to 1000 LPM) as well to assess the change in cutoff size over the concerned flow range. The cutoff varies between 2.56 μm to 2.49 μm in the considered flow range. Along with cutoff size, changes in the sharpness of the efficiency curves were observed as well. The σ_g values were poor at upper as well as lower limits of flow rates. During the field sampling with HVIA, the average operating flow rate range varied between 950 – 910 LPM. Therefore, on one hand, higher flow rate is not of much concern but the dropping air flow rate over the sampling duration could introduce difference in measurements compared to the actual concentrations because of the shift in cutoff size and change in efficiency curve sharpness. The flow rate of the sampler is expected to drop over long duration of sampling because of the resistance caused by particle-loaded filter. Therefore, the above exercise helps to understand the performance of the assembly in terms of cutoff over dropping flow rate.

Field Evaluation of HVIA

$PM_{2.5}$ Mass Concentration

A comparison of $PM_{2.5}$ mass concentrations measured using the 2 samplers for the sampling duration is shown in Fig. 6 with linear regression line and correlation coefficient. The fact that the measuring balance used to weigh A4 size filter (used with retrofit high volume sampler) could go down to 0.1 mg level as compared to the microbalance (used to weigh 47 mm Teflon filters used with low flow rate sampler), introduces an inherent cause for difference in the measured concentrations.

In addition, handling large A4 size filters is difficult and there could be minor loss of fiber, especially at the edges during loading and unloading of the filter (Kim et al., 2003). The filter face velocity for low flow rate sampler (~ 20 cm/s) was calculated to be approximately half of that for the high volume sampler (~ 36 cm/s). This suggests that the high volume sampler might lose some amount of semi-volatile fraction of $PM_{2.5}$ (as can be seen from WSOC concentrations discussed in later section) during sampling (Zhang and McMurry, 1992; Cheng and Tsai, 1997). The average operating flow rate of HVIA varied between 950 – 910 LPM over the whole sampling duration. Hence, the possible change in cutoff size and efficiency curve sharpness could also add to the discrepancy in mass concentrations obtained from the 2 samplers. Majority of the

$PM_{2.5}$ mass concentrations are observed to be on the lower side because of the monsoon season prevalent during the sampling schedule. Previous inter-comparison studies have largely relied on a good correlation coefficient even when actual gravimetric mass show large differences between two different samplers (Lehocky and Williams, 1996; Hill et al., 1999). High correlation (R^2 = 0.82) value established a good consensus between the two samplers (t-Test: $p < 0.05$).

The mean $PM_{2.5}$ mass concentration obtained from the high volume impactor and low flow rate sampler is 28.62 μg/m^3 and 33.35 μg/m^3, respectively. In one of the studies performed earlier during April-July 2011 (majorly during pre-monsoon season) at IIT Kanpur, reported a mean $PM_{2.5}$ mass concentration of 37.61 μg/m^3 (n = 36) (Ghosh et al., 2014). Since the PM sampling in current study was done in the monsoon season, lower values of mass concentrations were observed in this study.

$PM_{2.5}$ Elemental Concentrations

The elemental concentrations (Ca, Fe, Zn and Mg) obtained with $PM_{2.5}$ HVIA and low flow rate sampler are shown in Fig. 7, along with the linear regression line. Out of all the elements, Calcium and Iron followed by Zinc and Magnesium were observed to be in higher concentrations in the fine mode. Many of the sampling days were affected by monsoon rainfall due to which the concentrations of all the elements were very low and were contributed mainly by the local sources (including local industries, traffic and soil re-suspension). The mean concentration of Ca, Fe, Zn and Mg were 0.28 ± 0.11, 0.27 ± 0.08, 0.25 ± 0.09 and 0.23 ± 0.09 μg/m^3, respectively, for low flow sampler. The mean concentration of Ca, Fe, Zn and Mg were 0.24 ± 0.11, 0.23 ± 0.08, 0.17 ± 0.06 and 0.14 ± 0.06 μg/m^3, respectively, for HVIA. The Ca, Fe, Zn and Mg concentrations were correlated with R^2 values of 0.76, 0.80, 0.64 and 0.43, respectively, showing moderate to good consensus between the two samplers. Since only a portion of the filter is being used for chemical analysis, the correlations would strongly depend on how evenly the particles were collected over the filters. Such effect becomes more pronounced in case of species which are in low concentrations, like Zn and Mg.

$PM_{2.5}$ Anion Concentrations

The comparison between the 2 samplers for the $PM_{2.5}$ anionic concentrations are shown in Fig. 8. While sulphate and nitrate were the two dominant anionic species, chloride was also found in low concentrations. The mean concentration of sulphate, nitrate and chloride were 3.24 ± 1.48, 1.66 ± 0.83 and 0.69 ± 0.19 $\mu g/m^3$, respectively, for low flow sampler while those for HVIA were 2.88 ± 1.57, 1.40 ± 0.67 and 0.58 ± 0.20 $\mu g/m^3$, respectively. The sulphate and nitrate concentrations were correlated with R^2 values of 0.89 and 0.85, respectively. Ambient sulfate concentration is a stable chemical species and a good surrogate for ambient sources (Suh et al., 1993; Long et al., 2001). A sensitive detection method as ion chromatography is a far more superior compared to gravimetric analysis; therefore the sampler inter-comparison based on particulate sulfate and nitrate establishes better consensus between the information obtained with the 2 samplers.

$PM_{2.5}$ WSOC/WSTC Concentrations and Overall $PM_{2.5}$ Chemical Composition

The mean concentration of total water soluble carbon and WSOC were 2.46 ± 0.88 and 2.43 ± 0.89 $\mu g/m^3$, respectively, for low flow sampler while those for HVIA were 2.25 ± 0.75 and 2.22 ± 0.75 $\mu g/m^3$, respectively. The comparison between the 2 samplers for the $PM_{2.5}$ total water soluble carbon and WSOC concentrations are shown in Fig. 9. Strong correlations were obtained for total water soluble carbon (R^2 = 0.84) and WSOC (R^2 = 0.85) between the two samplers. Carbon concentrations are pretty low, especially in this part of the year, and are mostly contributed by vehicular emissions. WSIC contribution was found very low throughout the sampling period. Fig. 10 shows a comparison of the overall $PM_{2.5}$ chemical composition in terms of percentage, obtained from the 2 samplers.

CONCLUSIONS

This study involved the design and development of a multiple slit nozzle-based high volume $PM_{2.5}$ impactor assembly (HVIA) and validation of the same by comparing it with an existing low flow rate $PM_{2.5}$ impactor. The $PM_{2.5}$ mass concentration, elemental, anion and WSOC/WSIC concentration obtained after co-located sampling from the 2 samplers established good correlations between the two. The newly developed slit impactor assembly can be fitted to the existing PM_{10} high volume dust sampler quite comfortably, via replacing 4 screws and introducing appropriate amount of inert rubber gasket lining to provide leak free system. This is a major design improvement and makes it very attractive and user friendly. Particles of size greater than 10 µm are removed by the cyclone separator and only the finer fraction reaches the impactor. Therefore, particle loading over the impaction substrate was of least concern. Lower pressure drop across the multiple slit impactor as compared to other multiple round nozzle impactors at similar flow rate ensures that no change in the installed blower (in the dust sampler) or its settings is required. The whole system can comfortably be run under high dust loading conditions for long durations since now the filter is loaded with $PM_{2.5}$.

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REFERENCES

Alkuwairan, M.Y. (2012). Polygenetic dolomite in subtidal sediments of Northern Kuwait Bay, Kuwait. Dissertation, Colorado School of Mines.

Baron, P.A., and Willeke, K. (2001). *Aerosol measurement*. Willey & Sons Inc., New York.

Bhatnagar, A. (2011). *Environmental Cardiology: Pollution and Heart Disease*. Royal Society of Chemistry, Cambridge, UK.

Chakraborty, A. and Gupta, T. (2010). Chemical Characterization and Source Apportionment of Submicron (PM_1) Aerosol in Kanpur Region. *Aerosol Air Quality Res.* 10: 433-445.

Cheng, Y. and Tsai, C. (1997). Evaporation loss of ammonium nitrate particles during filter sampling. *J. Aerosol Sci.* 28: 1553–1567.

Cohen, A.J., Ross, A.H., Ostro, B., Pandey, K.D., Krzyzanowski, M., Künzli, N., Gutschmidt, K., Pope, A., Romieu, I., Samet, J.M. and Smith, K. (2005). The global burden of disease due to outdoor air pollution. *J. Toxicol. Environ. Health* 68: 1301-1307.

Deelman, J.C. (1999). Low-temperature nucleation of magnesite and dolomite. *J. Mineral. Geochem.* 7: 289-302.

Demokritou, P., Kavouras, I.G., Harrison, D. and Koutrakis, P. (2001). Development and evaluation of an impactor for a $PM_{2.5}$ speciation sampler. *J. Air Waste Manag. Assoc.* 51: 514-523.

Dockery, D.W. and Pope III, C.A. (1994). Acute effects of particulate air pollution. *Annu. Rev. Pub. Health* 15: 107-132.

Fang, C.P., Marple, V.A. and Rubow, K.L. (1991). Influence of cross-flow on particle collection characteristics of multi-nozzle impactors. *J. Aerosol Sci.* 22: 403-415.

Ghosh, S., Gupta, T., Rastogi, N., Gaur, A., Misra, A., Tripathi, S.N., Paul, D., Tare, V., Prakash, O., Bhattu, D., Dwivedi, A.K., Kaul, D.S., Dalai, R. and Mishra, S.K. (2014). Chemical Characterization of Summertime Dust Events at Kanpur: Insight into the Sources and Level of Mixing with Anthropogenic Emissions. *Aerosol Air Quality Res.* 14: 879-891.

Gupta, T., Chakraborty, A. and Ujinwal, K.K. (2010). Development and Performance Evaluation of an Indigenously Developed Air Sampler Designed to Collect Submicron Aerosol. *Annals of the Indian National Academy of Engineering (INAE)* VII: 189-193.

Gupta, T., Jaiprakash, and Dubey, S. (2011). Field performance evaluation of a newly developed $PM_{2.5}$ sampler at IIT Kanpur. *Sci. Total Environ.* 409: 3500-3507.

Gupta, T. and Mandaria, A. (2013). Sources of Submicron Aerosol during Fog Dominated Wintertime at Kanpur. *Environ. Sci. Pollut. Res.* 20: 5615-5629.

Harrison, R.M. and Yin, J. (2000). Particulate matter in the atmosphere: which particle properties are important for its effects on health? *Sci. Total Environ.* 249: 85-101.

Hering, S. (1995). Impactors, cyclones and other inertial and gravitational collectors, In *Air Sampling Instruments for Evaluation of Atmospheric Contaminants*, (eds.), B.S. Cohen and S.V. Hering, ACGIH, Cincinnati, p. 315-375.

Hill, J.S., Patel, P.D. and Turner, J.R. (1999). Performance characterization of the MiniVol $PM_{2.5}$ sampler. *In Proceedings of 92nd Annual Meeting of Air & Waste Management Association*, St. Louis, MO, June 20–24, paper 99–617.

Hinds, W.C. (1999). *Aerosol Technology*. John Willey and Sons Inc, New York.

Huang, Y.C. and Ghio, A.J. (2006). Vascular effects of ambient pollutant particles and metals. *Curr. Vasc. Pharmacol.,* 4: 199-203.

Kim, E., Hopke, P.K. and Edgerton, E.S. (2003). Source identification of Atlanta aerosol by positive matrix factorization. *J. Air Waste Manage. Assoc.* 53: 731–739.

Korrick, S.A. (1998). Effects of ozone and other pollutants on the pulmonary function of adult hikers. *Environ. Health Persp.* 106: 93-99.

Kumar, A. and Gupta, T. (2014). Development and Laboratory Performance Evaluation of a Variable Configuration PM1/PM2.5 Impaction-Based Sampler. *Aerosol Air Quality Res.* doi: 10.4209/aaqr.2014.11.0307.

Kunzli, N. and Tager, I.B. (2005). Air pollution from lung to heart. *Swiss Med. Wkly.* 135: 697–702.

Lehocky, A. and Williams, P.L. (1996). Comparison of respirable samplers to direct-reading realtime aerosol monitors for measuring coal dust. *Am. Indl. Hyg. Assoc.* 57: 1013–1018.

Long, C.M, Suh, H.H., Catalano, P.J. and Koutrakis, P. (2001). Using time and size-resolved particulate data to quantify indoor penetration and deposition behavior. *Environ. Sci. Tech.* 35: 2089–2099.

Marple, V.A., Rubow, K.L. and Behm, S.M. (1991). A micro-orifice uniform deposit impactor (MOUDI). *Aerosol Sci. Tech.* 14: 434-441.

Marple, V.A. and Willeke, K. (1976). Inertial Impactors: Theory, Design and Use, In *Fine Particles: Aerosol Generation, Measurement, Sampling and Analysis*, B.Y.H. Liu, (eds.) Academic Press, New York, p. 411-446.

National Research Council (1998). Research Priorities for Airborne Particulate Matter: Immediate Priorities and a Long-Range Research Portfolio. Washington DC, National Academy Press.

Neas, L.M. (1999). A case-crossover analysis of air pollution and mortality in Philadelphia. *Environ. Health Persp.* 107: 629-631.

Park, S., Liu, H. and Rubow, K. (1992). Effect of coating thickness on particle bounce in inertial impactors. *Aerosol Sci. Tech.* 16: 141-150.

Peters, M.T., Vanderpool, R.W. and Wiener, R.W. (2001). Design and Calibration of the EPA $PM_{2.5}$ Well Impactor Ninety-Six (WINS). *Aerosol Sci. Tech.* 34: 389-397.

Pope, C.A. III, Burnett, R.T., Turner, M.C., Cohen, A., Krewski, D., Jerrett, M., Gapstur, S.M. and Thun, M.J. (2011). Cardiovascular mortality and exposure to airborne fine particulate matter and cigarette smoke: shape of the exposure-response relationship. *Environ. Health Persp.* 119: 1616-1621.

Rao A.K. (1975). An experimental study of inertial impactors. Ph.D. Dissertation, ParticIe Technology Laboratory, Mechanical Engineering Dept., University of Minnesota.

Schwartz, J. and Dockery, D.W. (1992). Increased Mortality in Philadelphia Associated with Daily Air Pollution Concentrations. *Am. Rev. Respir. Dis.* 145: 600-604.

Schwartz, J., Dockery, D.W. and Neas, J.M. (1996). Is daily mortality associated specifically with fine particles? *J. Air Waste Manage. Assoc.* 46: 927-939.

Sharma, R.K. and Agrawal, M. (2005). Biological effects of heavy metals: an overview. *J. Environ. Biol.* 26: 301-313.

Stone, E.A., Yoon, S. and Schauer, J. (2011). Chemical Characterization of Fine and Coarse Particles in Gosan, Korea during Springtime Dust Events. *Aerosol Air Quality Res.* 11: 31-43.

Suh, H.H., Koutrakis, P. and Spengler, J.D. (1993). Validation of personal exposure models for sulfate and aerosol strong acidity. *J. Air Waste Manage. Assoc.* 43: 845–50.

Turner, J.R. and Hering, S.V. (1987). Greased and oiled substrates as bounce free impaction surfaces. *J. Aerosol Sci.* 18: 215-224.

Zhang, X. and McMurry, P.H. (1992). Evaporation losses offine particulate nitrates during sampling. *Atmos. Environ.* 26: 3305–3312.

Web Resources:

WHO (2006). Air quality guidelines: Global update 2005. Geneva. *(http://www. euro.who.int/)*

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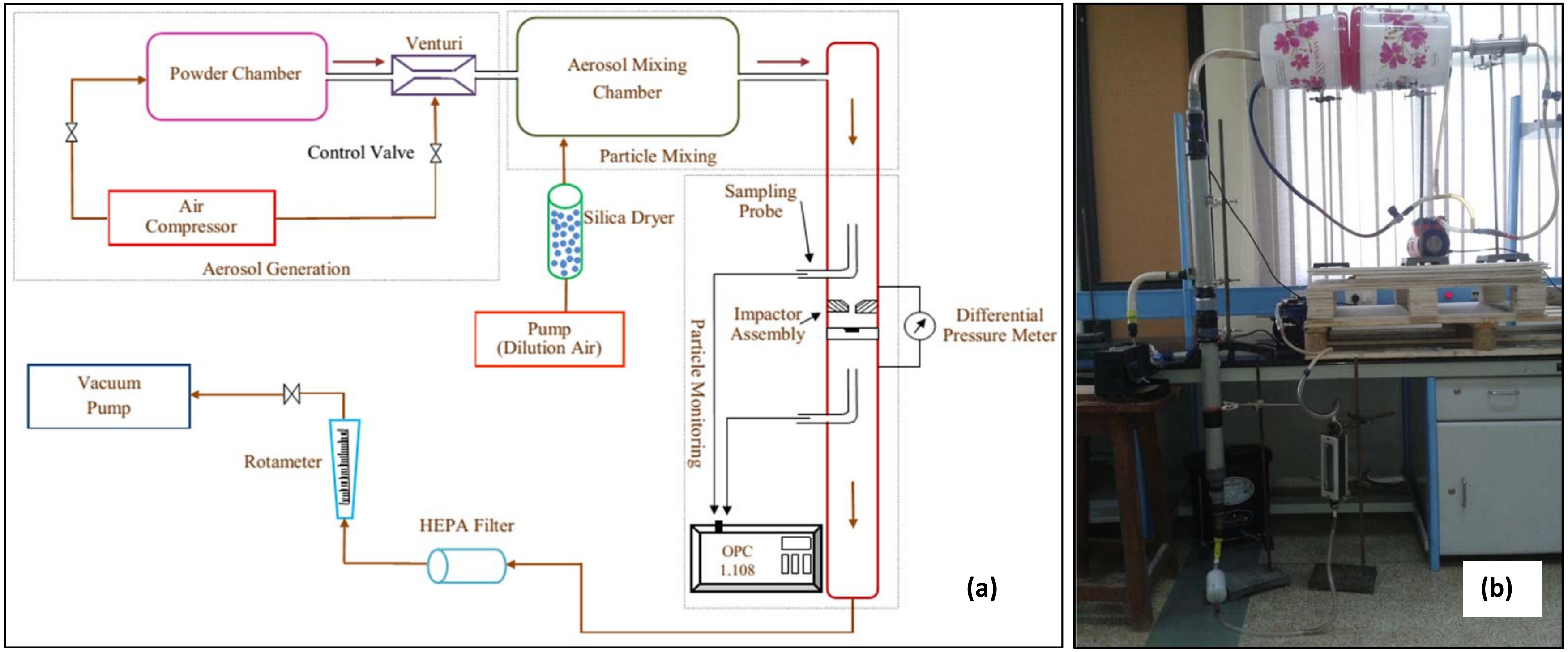

Fig. 1(a). Schematic of the Experimental Setup for Impactor Characterization in the Lab, **(b)** Laboratory Testing Rig.

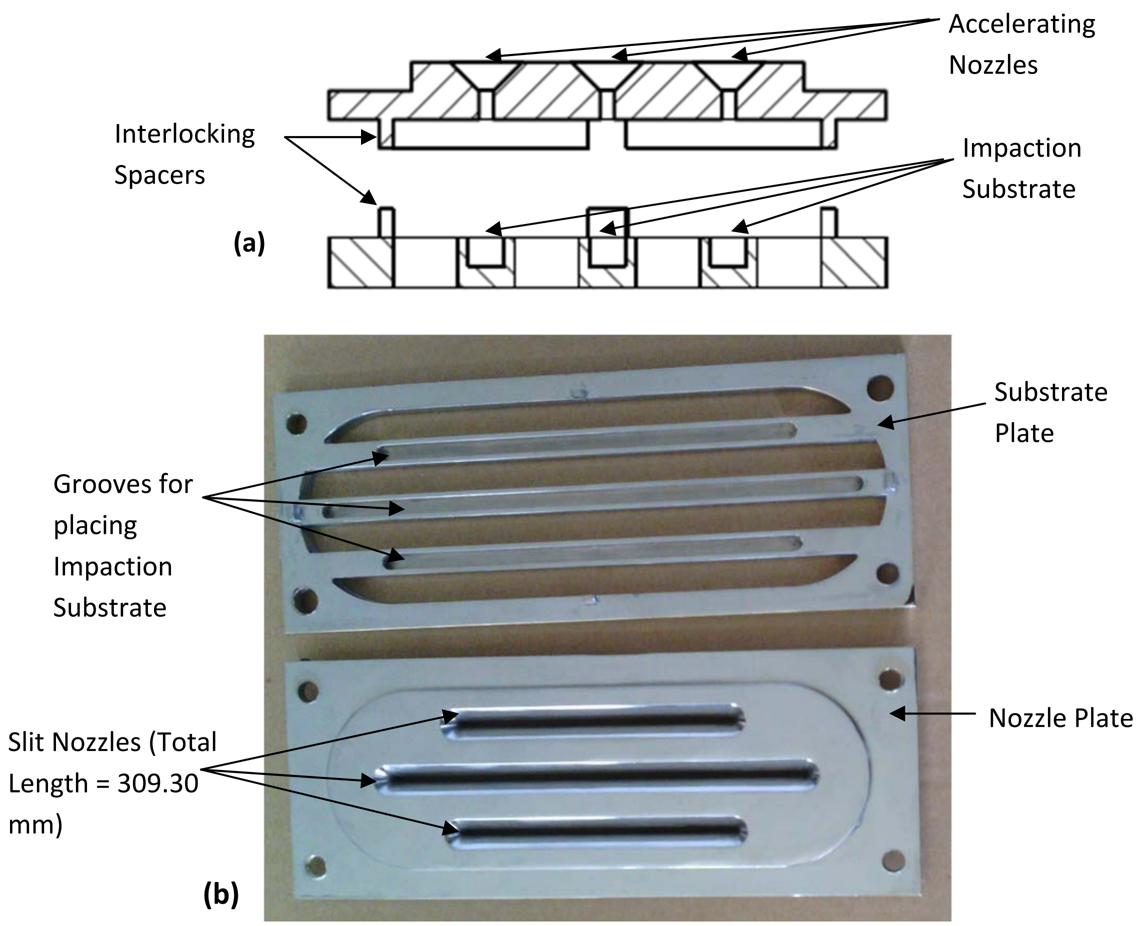

Fig. 2(a). Schematic of the novel slit nozzle-based high volume impactor assembly, **(b)** Components of the final impactor assembly.

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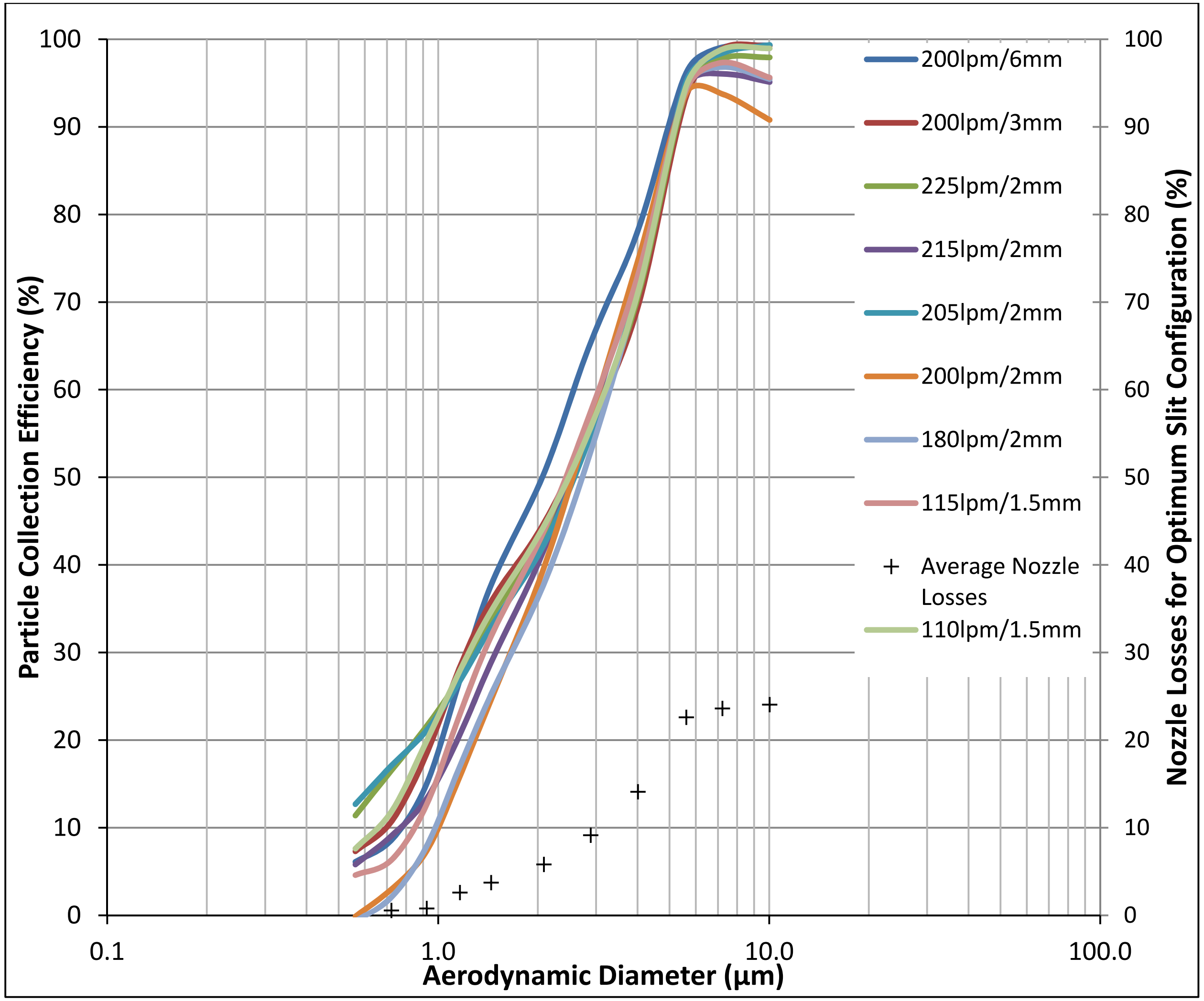

Fig. 3. Collection Efficiency for Different Slits and Nozzle Losses for Optimum Slit Nozzle Configuration.

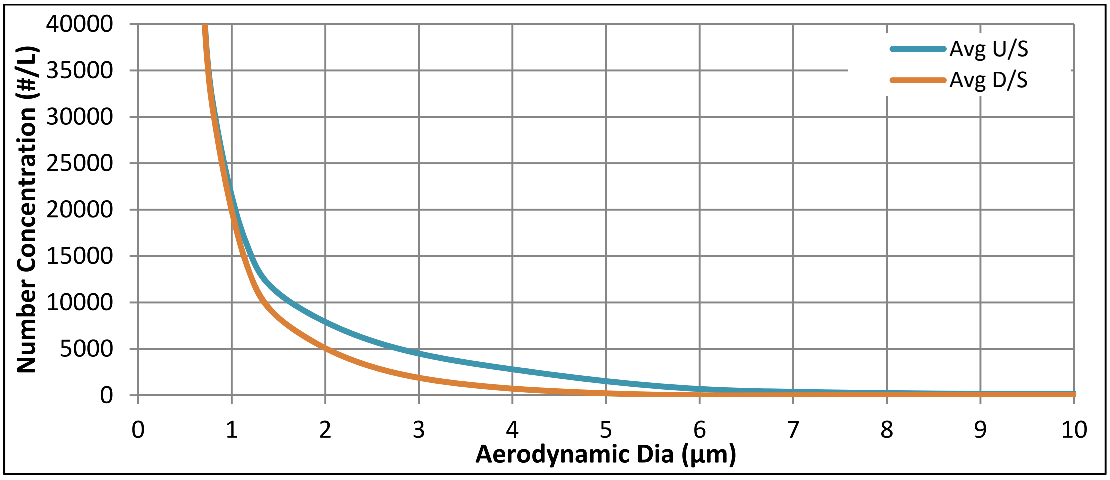

Fig. 4. Average aerosol number concentration on the upstream (U/S) and downstream (D/S) of impactor in a parameterization test at 215 LPM.

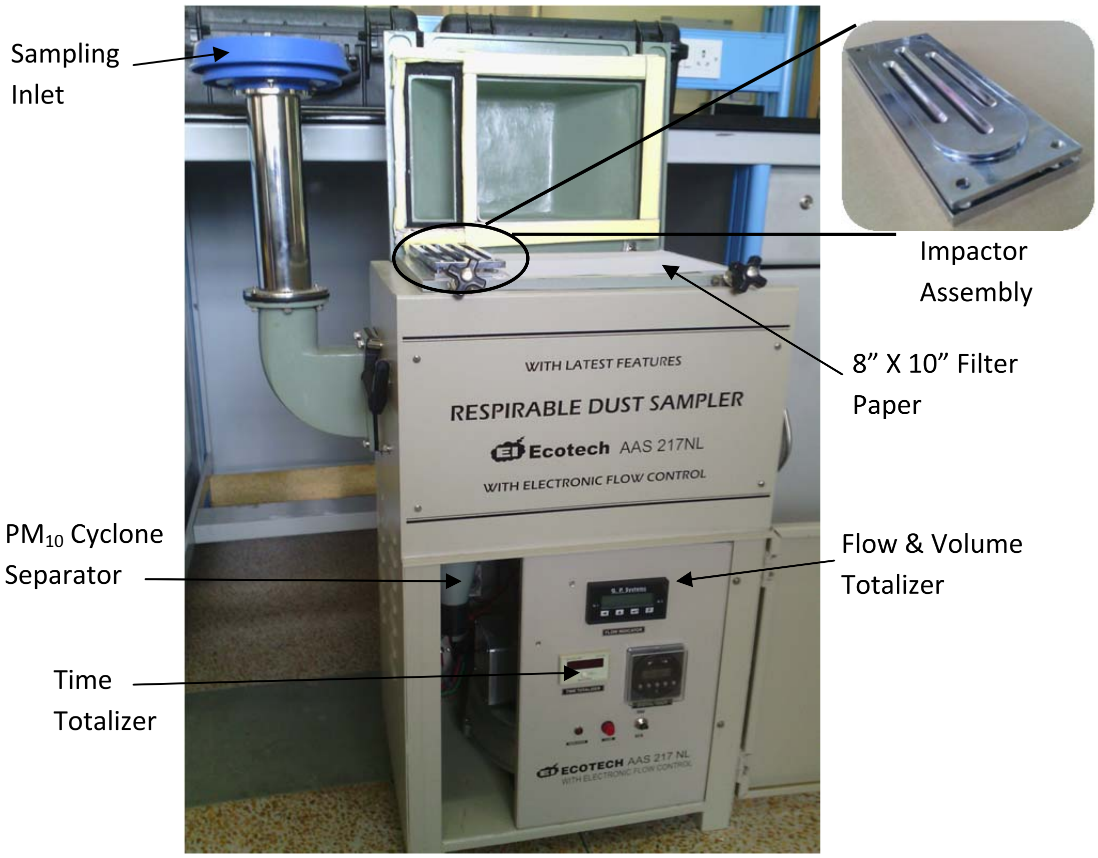

Fig. 5. Components of the High Volume Dust Sampler with the Novel Slit Impactor Assembly (HVIA) installed on it.

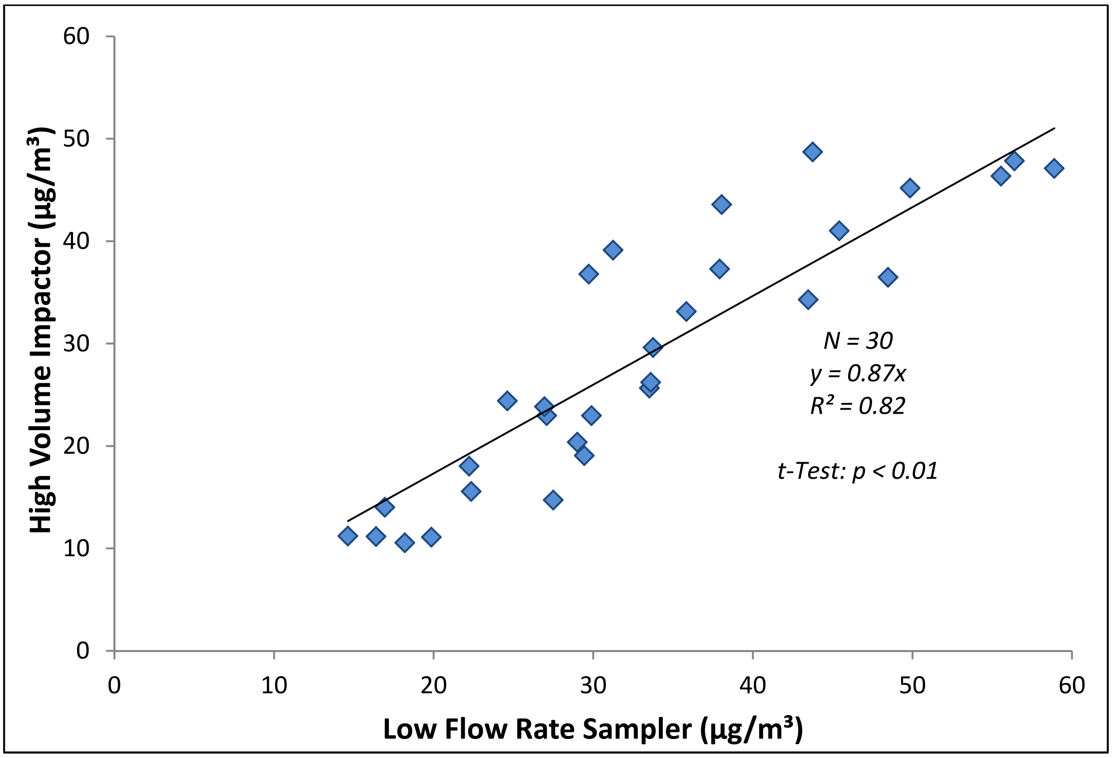

Fig. 6. Comparison of $PM_{2.5}$ Mass Concentration measured with HVIA and Low Flow Rate Impactor.

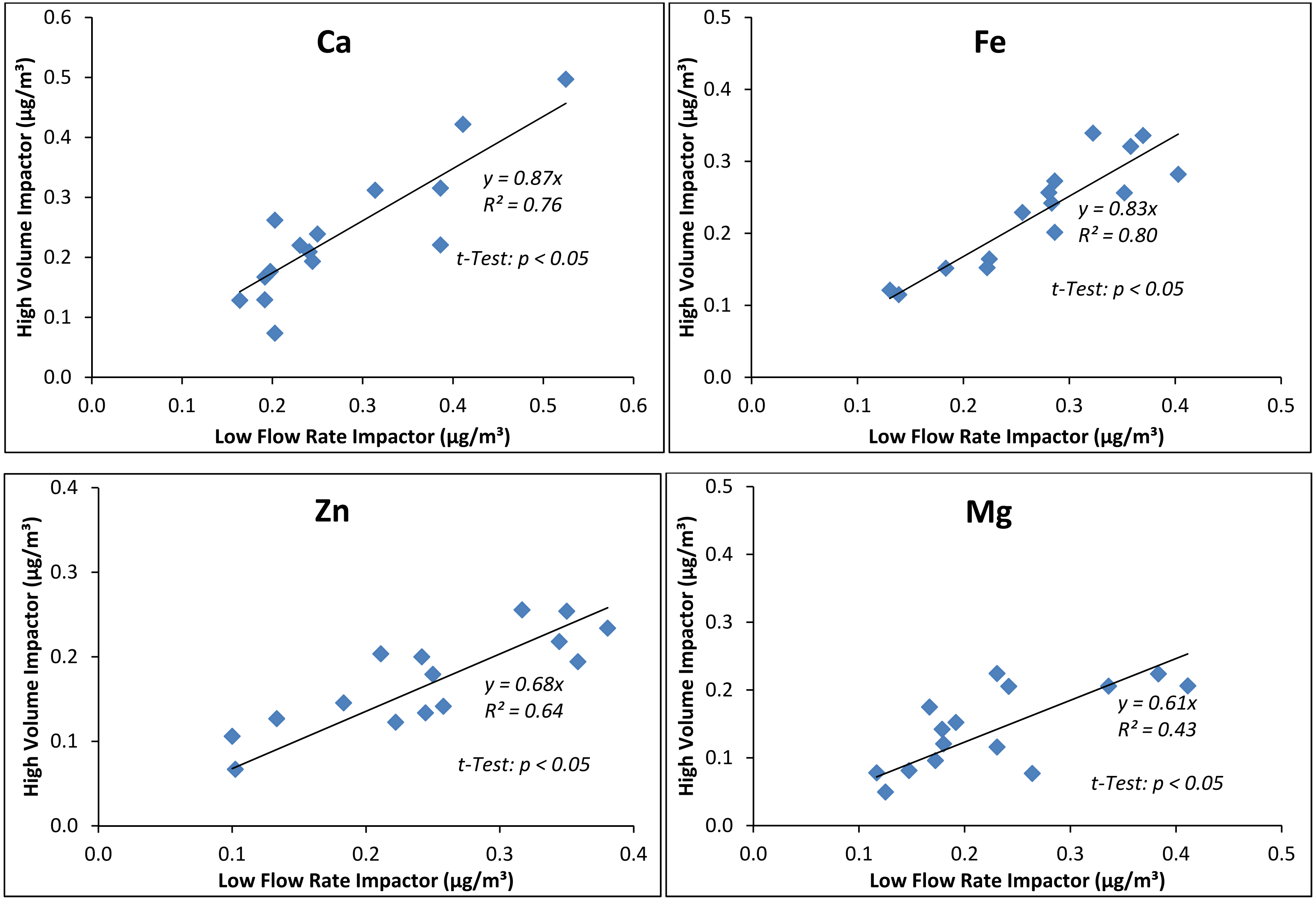

Fig. 7. Comparison of elemental concentrations measured with HVIA and Low Flow Rate Impactor.

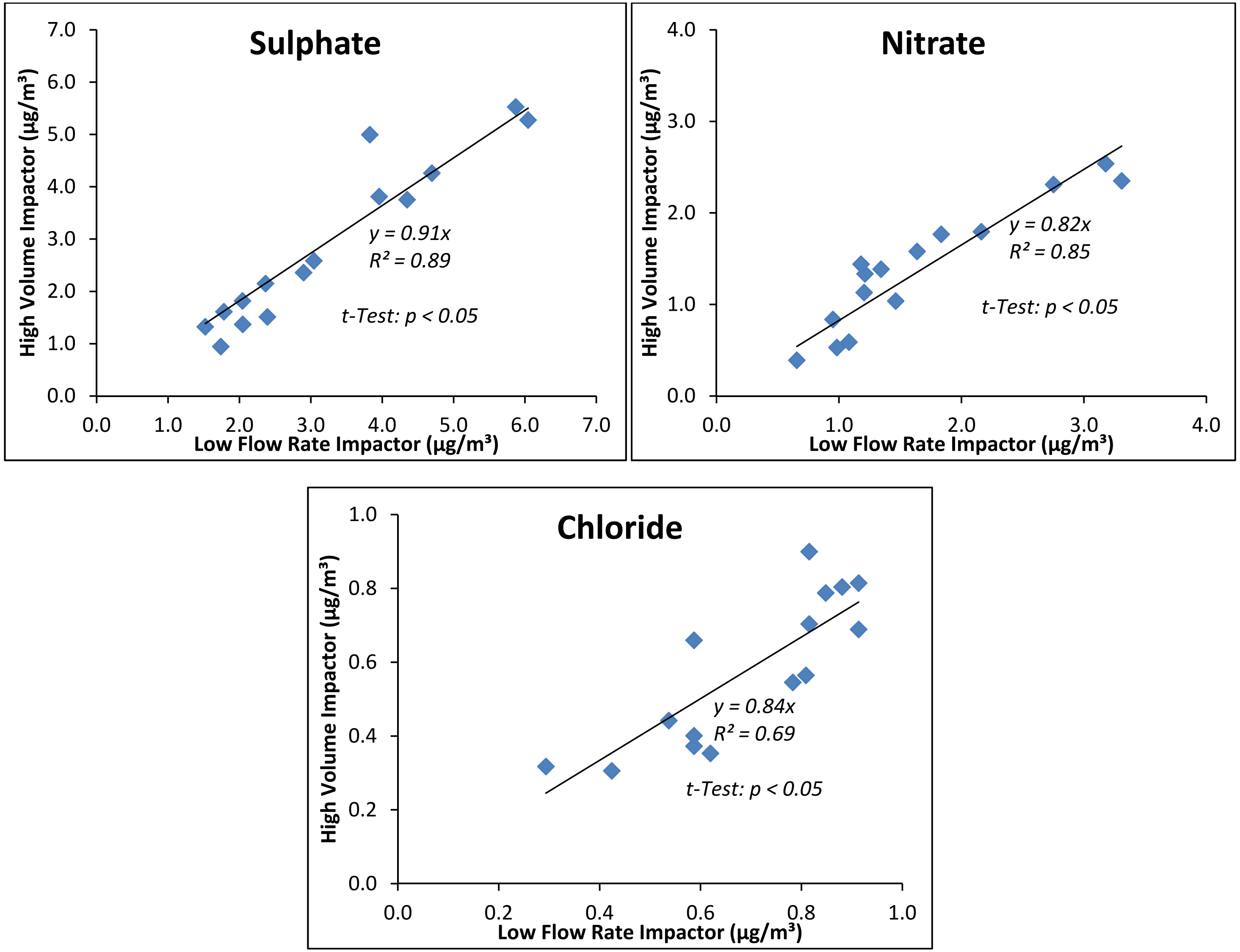

Fig. 8. Comparison of anion concentrations measured with HVIA and Low Flow Rate Impactor.

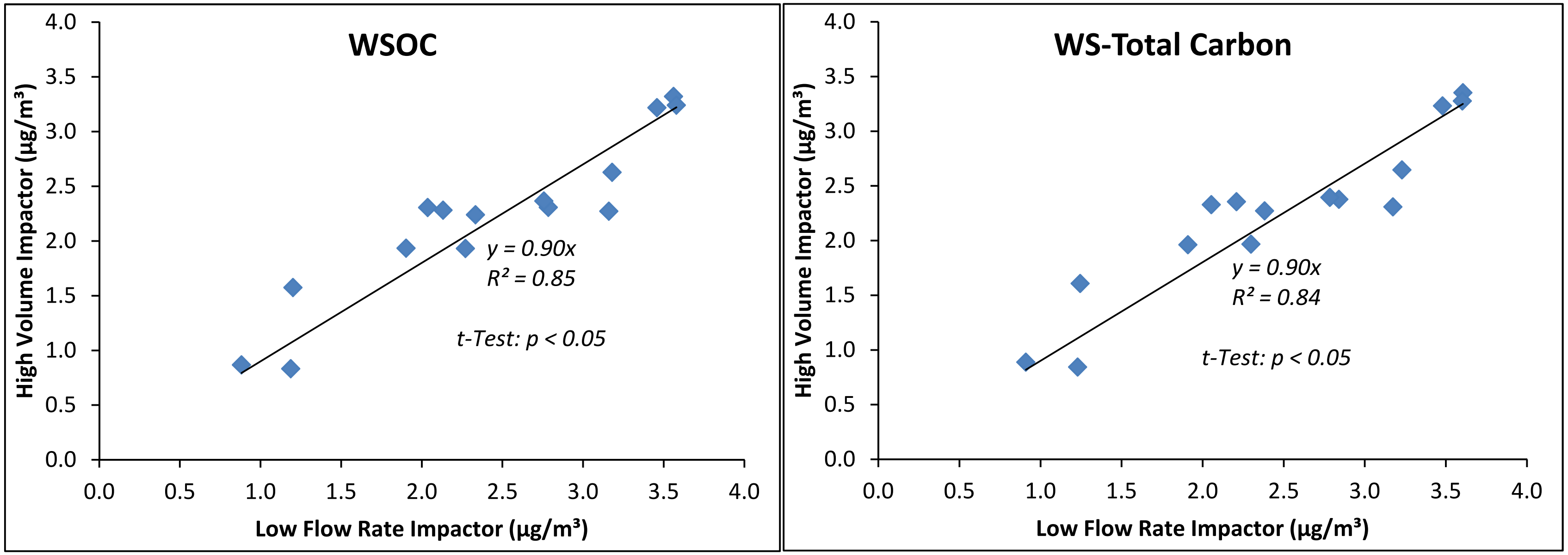

Fig. 9. Comparison of WSOC and WSTC concentrations measured with HVIA and Low Flow Rate Impactor.

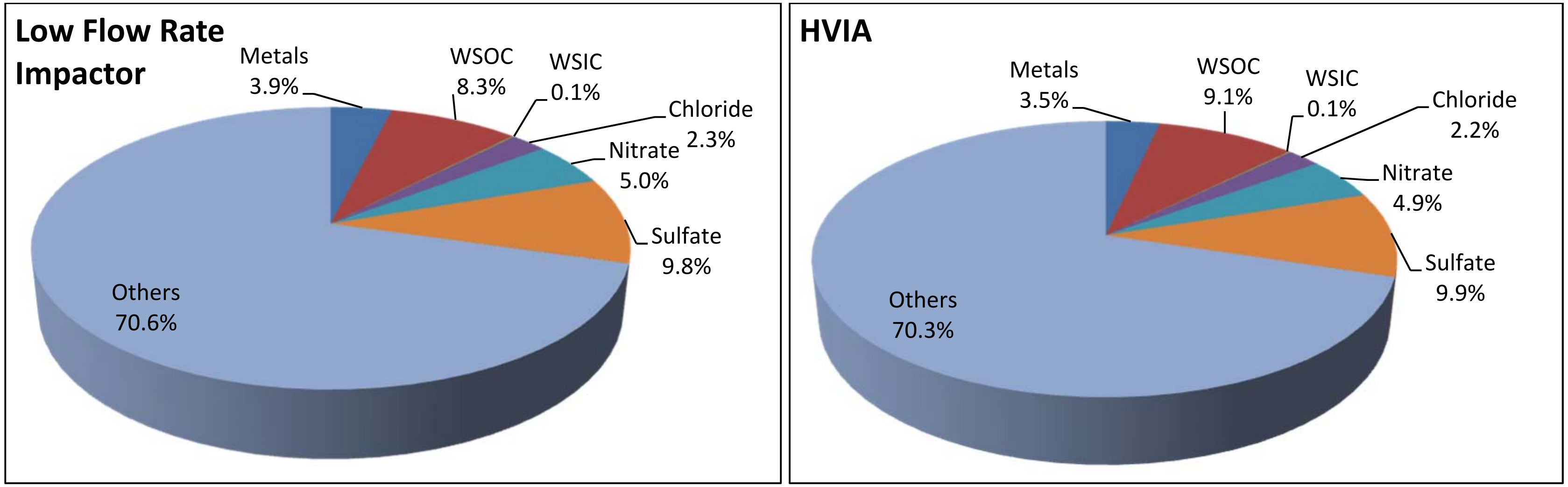

Fig. 10. Comparison of $PM_{2.5}$ chemical composition obtained from HVIA and Low Flow Rate Impactor.

List of Tables

Table 1. Results of the Initial Parametric Investigation for Multiple Slit-based Nozzles

Nozzle Width (mm)	Q (LPM)	S/W	Number of sample runs	$d_{50\ exp}$ (µm)	$\sqrt{Stk_{50exp}}$	σ_g	ΔP (kPa)	Re	Shape of Efficiency Curve
1.5	110	2.0	2	2.48	0.66	2.39	0.17	3459	Moderate
1.5	115	2.0	2	2.43	0.66	2.18	0.20	3616	Moderate
2	180	2.0	2	2.69	0.70	2.34	0.27	5789	Moderate
2	200	2.0	2	2.61	0.71	2.19	0.34	6432	Moderate
2	205	2.0	3	2.56	0.71	2.65	0.34	6593	Poor
2	215	2.0	5	2.51	0.71	2.14	0.35	6915	Moderate
2	225	2.0	2	2.49	0.73	2.63	0.37	7236	Poor
3	200	2.0	2	2.45	0.67	2.39	0.75	14559	Moderate
6	200	2.0	2	2.08	0.37	2.17	-	24266	Moderate

Supplementary Information

Development and Field Evaluation of a Multiple Slit Nozzle-Based High Volume $PM_{2.5}$ Inertial Impactor Assembly (HVIA)

Anand Kumar[a] and Tarun Gupta[a,b]

[a]*Environmental Engineering and Management Program*

Indian Institute of Technology Kanpur, Kanpur - 208016 (India)

[b]***Corresponding author, Tel: +91-512-2597128; E-mail address: tarun@iitk.ac.in***

TABLES

Table S1. Sampling Schedule for testing Dust Sampler retrofitted with novel Multiple Slit Nozzle-based High Volume $PM_{2.5}$ Impactor Assembly (HVIA)

Sampler Type	Aerosol Type	Sampling Site	Sampling Period	Sampling Duration	Number of Samples
Low Flow Sampler	$PM_{2.5}$	CESE, IIT Kanpur	09:00 AM to 05:00 PM	8 hours	30
HVIA	$PM_{2.5}$	CESE, IIT Kanpur	09:00 AM to 05:00 PM	8 hours	30

FIGURES

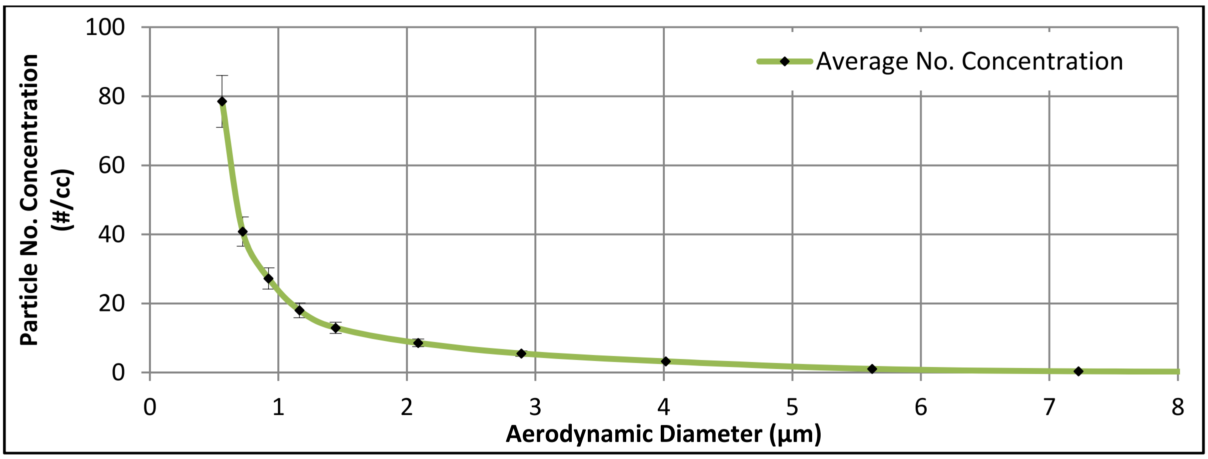

Fig. S1. Size Distribution Plot showing stability over 10 representative sample runs (mean concentration ± one SD) of Aerosol Generation system.

Supplementary Information

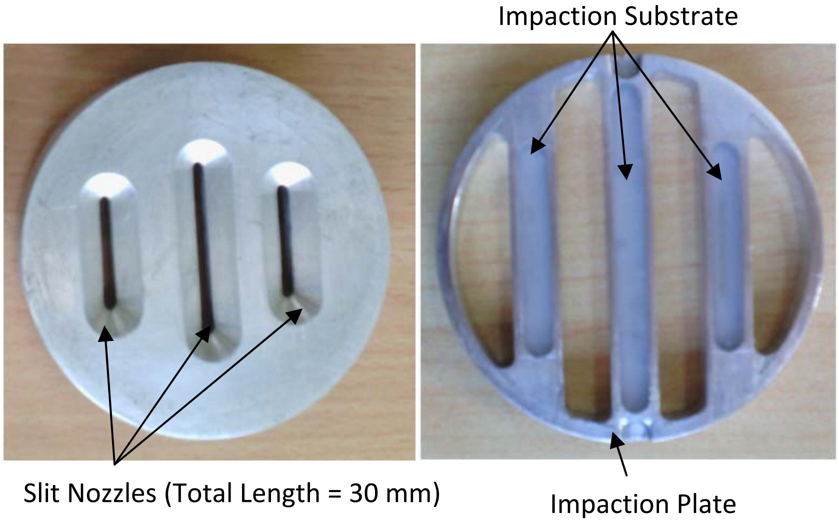

Fig. S2. Components of the Optimum Multiple-Slit Nozzle Configuration operating at a Medium Flow Rate (Q = 215 LPM, d_{50} = 2.51 µm, $\sqrt{Stk_{50exp}}$ = 0.71).

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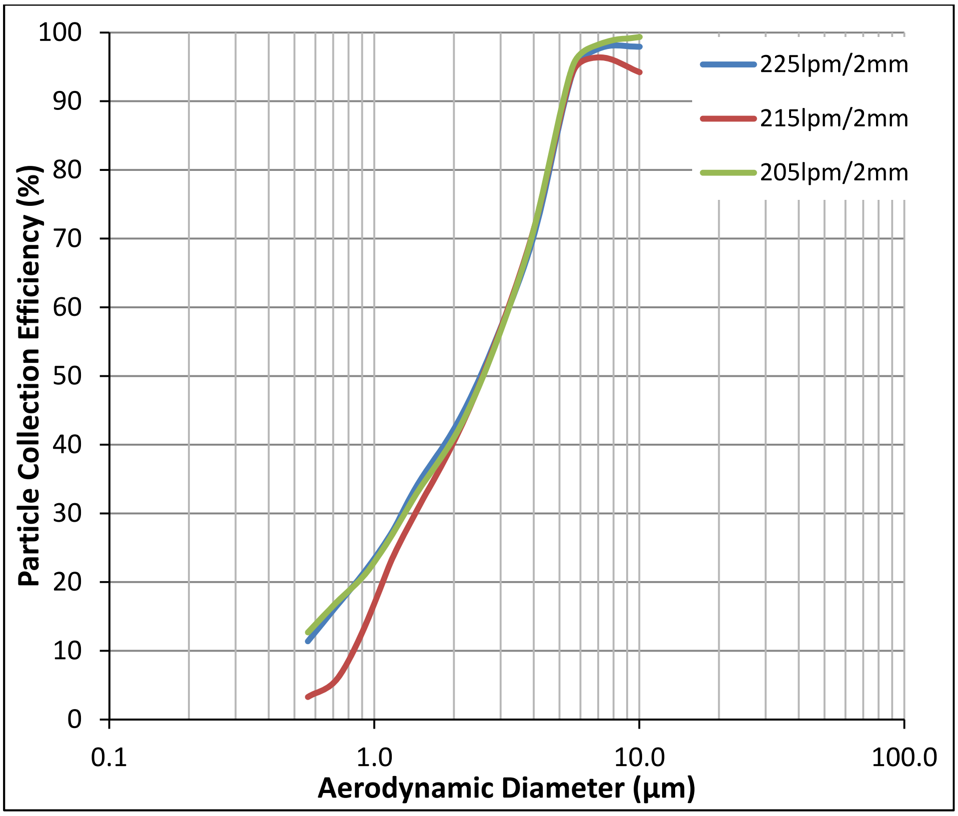

Fig. S3. Comparison of Collection Efficiency of Medium Flow Slit Nozzle over operating flow range of High Volume Dust Sampler.